DealLens 2.0: From Evidence to Agentic Action

From Pivotal Evidence to Expert Memory, Reasoning Audits, and Actionable Professional Artifacts

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Starting point: From DealLens to accountable AI action

What if a sales intelligence system could do more than predict which deals are at risk: could it reveal the moment a deal changed, learn why an expert disagrees, and help shape the next move? DealLens provides a concrete starting point. Combining Salesforce opportunity outcomes with Gong conversations, it reconstructs cumulative deal trajectories from Call 4 onward, surfaces influential signals and source evidence, and offers revisable guidance. Its predictive layer identifies 82.5% of eventual losses by Call 4 and 93.2% by Call 8. Rather than treating these early warnings as the endpoint, this agenda asks how an evidence-grounded system can progress from detecting change to acquiring expertise, auditing judgment, and producing trustworthy, forward-looking artifacts that professionals can inspect and approve. While DealLens serves as the initial research setting, these challenges extend to other evidence-intensive domains, such as healthcare, financial services, and education, where professionals must interpret fragmented information, exercise expert judgment, and make consequential decisions.

Research gap and central question

Conversation research increasingly treats dialogue as a trajectory rather than a collection of isolated utterances. Work on pivotal moments identifies consequential changes within conversations [1], while longitudinal studies examine professional adaptation after difficult episodes [2]. Yet DealLens currently updates risk after completed calls. A finer question is which turn introduces evidence that changes the current assessment—and whether professionals find that change consequential.

A second gap concerns how expertise is represented and evaluated. Canvil and teachable-feed research demonstrate interactive adaptation by non-ML experts [3,4], while Synthetic Cognitive Walkthrough compares human and AI task reasoning to reveal mismatches [5]. Feedback collection and reasoning audit are often treated separately. An accountable agent should be able to elicit tacit context, preserve approved knowledge with explicit scope and provenance, and separately assess where AI and professionals interpret identical evidence differently—and whether those differences affect decisions.

A third gap lies between interpreting evidence and supporting work that changes what happens next. Studies of contextualized research artifacts [6,7], PaperTok [8], and PosterMate [9] illuminate evidence translation and audience-aware generation. Professional action also requires credible strategies, tailored artifacts, organizational constraints, clarification when evidence is insufficient, and verifiable claim-to-source links. These needs motivate four distinct but connected studies spanning detection, learning, audit, and forward-looking action.

Together, these gaps point to a broader challenge: how can AI move beyond identifying risks and generating isolated recommendations toward becoming an adaptive, evidence-grounded partner in human decision-making? This motivates the central research question: How can conversational generative AI agents support evidence-based professional work by detecting consequential evidence shifts, learning from situated human expertise, auditing human–AI reasoning differences, and transforming validated judgments into traceable, context-specific actions?

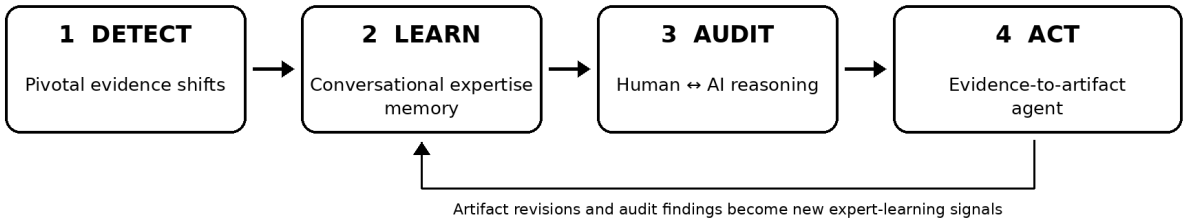


Figure 1. Research progression: detect consequential evidence shifts; learn scoped expertise; audit human–AI reasoning independently; and support evidence-grounded professional action. Findings and artifact revisions may inform future learning.

Four research questions

RQ	Focus	Research question
1	Pivotal evidence shifts	How can AI identify turn-level conversational moments that meaningfully change an evidence-based assessment of deal trajectory, and which of those shifts professionals judge consequential?

RQ	Focus	Research question
2	Conversational expertise learning	How can a conversational agent actively elicit, represent, and reuse tacit professional knowledge to adapt AI reasoning over time?
3	Reasoning divergence and consequences	Where, why, and with what consequences do AI systems and experienced professionals diverge in evidence selection, interpretation, counterevidence, and action reasoning at the same pivotal moment?
4	Evidence-to-action support	How can an evidence-grounded agent transform validated professional judgments into context-specific strategies and actionable work artifacts while preserving provenance, organizational constraints, and human control?

Study 1 — Detecting pivotal evidence shifts inside a customer call

RQ1 — How can AI identify turn-level conversational moments that meaningfully change an evidence-based assessment of deal trajectory, and which of those shifts professionals judge consequential?

Research purpose

DealLens already provides evidence-based early warning at the call level by recomputing cumulative Loss risk after each completed call. Study 1 moves one level deeper: it asks which turn inside a call most changes the evidence-based assessment of the deal trajectory. A pivotal moment is therefore a buyer or professional turn after which predicted Loss risk changes substantially, either for better or for worse. The contribution is finer temporal resolution for professional sensemaking, not a causal claim that one utterance produced the final outcome.

Evidence-based turn-level detection

Each transcript is segmented into ordered conversational turns while preserving the evidence context DealLens already uses: prior calls, CRM state, deal stage, stakeholder role, and accumulated interaction history. A turn-level AI forecaster f_θ estimates Loss probability after every turn t from the history H_t. The call-level DealLens score provides the starting state when a call begins; the turn-level system then traces how each new piece of conversational evidence changes that assessment. This creates an evidence trajectory inside the call rather than only between calls.

Pivotal-moment definition

Let $p_t = f_\theta(H_t)$ denote predicted Loss probability after turn t. The local evidence shift and pivotality score are:

$$\Delta_t = p_t - p_{t-1}, \qquad PIV_t = |\Delta_t|$$

The sign of Δ_t preserves direction: $\Delta_t > 0$ means the new turn shifts the evidence toward a worse trajectory, whereas $\Delta_t < 0$ means improvement. PIV_t measures the magnitude regardless of direction. A turn is flagged when PIV_t exceeds a development-set threshold τ or ranks among the largest within-call shifts. This inspectable definition allows professionals to examine which new evidence changed the AI assessment.

Evaluation

Evaluation has three layers:

First, detection validity compares AI-ranked moments against independent annotations from blinded experienced SEs and sales managers, who identify consequential turns and their direction of change. Metrics include precision, recall, nDCG, ranking agreement, and inter-rater agreement. Four baselines test whether PIV captures more than conventional signals: (1) current risk level, selecting the highest-risk turns rather than the largest changes; (2) sentiment change, selecting the largest emotional shifts; (3) objection/topic flags, identifying predefined concerns without assessing their contextual significance; and (4) random selection, providing a reference. All methods use the same calls and selection budgets, ranking turns by their respective signals and comparing their top-ranked selections against expert annotations using identical metrics. This comparison tests whether detecting changes in evidence provides greater alignment with professional judgment than relying on risk severity, emotional shifts, or predefined conversational cues.

Second, explanation quality evaluates whether AI explanations are grounded in source evidence, contextually plausible, and free from important evidence omissions. Experts assess these dimensions independently to distinguish meaningful interpretation from unsupported inference.

Third, consequentiality and review usefulness tests whether detected moments deserve professional attention. Experts blindly evaluate three groups: high-PIV turns (substantial changes), high-risk but low-PIV turns (concerning states without meaningful changes), and random turns. They rate how substantially each moment changes their deal assessment and how useful highlighting it would be for review or next-action preparation. This tests whether surfacing what changed is more valuable than highlighting what is already risky.

Analysis

A sequential explanatory mixed-methods approach examines both whether PIV aligns with professional judgment and why that alignment varies across evidence types.

Quantitatively, mixed-effects regression tests associations between PIV and expert-rated consequentiality and review usefulness, accounting for participant, deal, deal stage, conversation position, and current Loss risk. This establishes whether PIV provides additional information beyond existing risk assessments and identifies cases of agreement and disagreement.

Qualitatively, researchers code representative agreement and disagreement cases into evidence-shift categories, such as technical objections, stakeholder uncertainty, competitive threats, timeline ambiguity, trust challenges, commitments, and next-step refusals. These categories are then connected to detection performance: high-PIV moments rated inconsequential may reflect overreaction to routine delays, while low-PIV moments rated consequential may reveal overlooked stakeholder dynamics. Integrating both analyses identifies which evidence shifts AI reliably recognizes, where it diverges from professionals, and why, establishing the contextual strengths and limitations of pivotal-moment detection.

Expected contribution

Study 1 extends DealLens from "this deal became riskier after Call 5" to "this specific turn inside Call 5 most changed the evidence-based assessment of its trajectory." It contributes a validated turn-level detector, evidence of professional consequentiality and review usefulness, and a taxonomy explaining where AI and expert judgments converge or diverge. The resulting cases can support the expertise learning of Study 2 and independent reasoning audits of Study 3.

Study 2 — Building a conversational expertise-learning agent

RQ2 — How can a conversational agent actively elicit, represent, and reuse tacit professional knowledge to adapt AI reasoning over time?

Research purpose

The planned DealLens interface allows professionals to inspect, edit, or reject AI guidance. Study 2 transforms these one-time corrections into a continuous expertise-learning mechanism. Rather than simply responding to feedback, the agent actively investigates disagreements, elicits missing professional context, and converts validated insights into reusable knowledge. A central research focus is how an agent can develop and maintain an explicit, inspectable memory of professional expertise, enabling it to improve reasoning across cases without requiring continuous retraining. This shifts the interaction from correcting individual AI responses toward progressively building a human-governed knowledge base that supports context-sensitive adaptation.

Agent loop and feedback conditions

The agent operates through three integrated capabilities: diagnosing disagreement, eliciting expertise, and applying validated knowledge. When professionals challenge an assessment, the agent identifies potentially missing context and asks targeted follow-up questions to understand their reasoning. It then proposes a knowledge update for professional review, distinguishing case-specific exceptions from broader heuristics. Once approved, the knowledge becomes available for relevant future cases, where the agent applies it selectively and checks whether it remains appropriate. The study compares three feedback conditions: (1) simple correction, allowing users to reject or edit AI guidance; (2) structured feedback, specifying whether the issue concerns evidence, interpretation, or recommended action; and (3) conversational elicitation, where the agent proactively explores the professional reasoning behind a disagreement.

Knowledge transfer, representation, and adaptation

Expert memory is the central mechanism connecting conversational learning to future agent behavior. Rather than storing raw feedback or automatically fine-tuning the underlying AI, the agent transforms confirmed insights into structured memory units containing evidence, interpretation, applicability scope, confidence, provenance, and expiry conditions. For example, a legal delay may be recorded as an account-specific exception, whereas the withdrawal of an influential stakeholder may become a broader professional heuristic. When encountering a new case, the agent retrieves relevant expert memories to inform its interpretation and recommendations. Crucially, memory remains inspectable and editable: professionals can examine which prior insights influenced a decision, correct inappropriate generalizations, or revoke outdated knowledge. The research therefore examines how memory-augmented agents can transfer tacit expertise across situations while preserving contextual boundaries and human control.

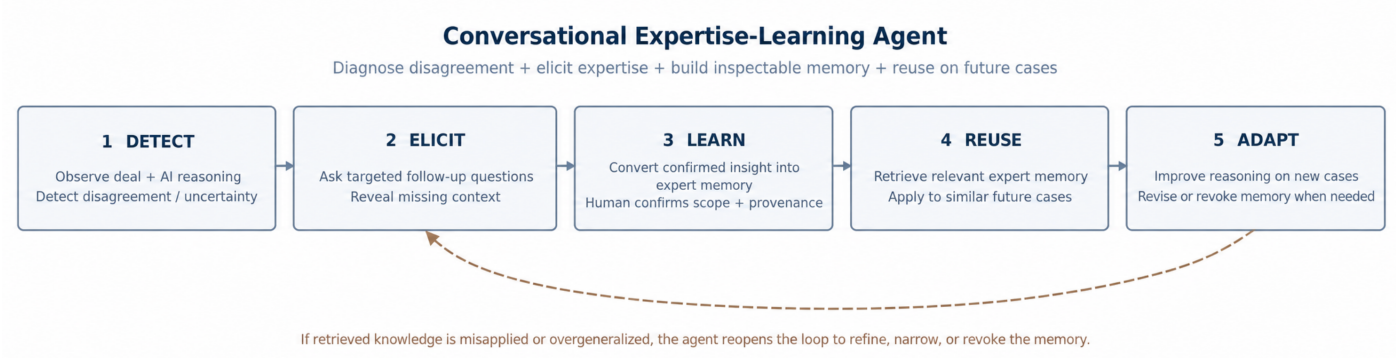


Figure 2. Conversational expertise-learning agent loop. Expert feedback becomes a human-approved memory update that can be retrieved, applied, monitored, and revised across future cases.

Evaluation

Evaluation proceeds in two stages. A 10–15 person pilot refines the agent's clarification questions, memory extraction, case difficulty, and experimental measures, informing sample-size planning. A subsequent counterbalanced within-subject study, provisionally involving 20–30 SEs and managers subject to power analysis, compares simple correction,

structured feedback, and conversational elicitation. Participants teach the agent on initial cases, after which its reasoning is evaluated before and after memory-based adaptation on held-out cases. These include structurally similar cases testing successful knowledge transfer and boundary cases testing whether account-specific or contextual knowledge is incorrectly generalized. The study examines whether conversational elicitation improves reasoning on new cases while preserving appropriate memory scope and professional control.

Analysis

The primary outcome is improvement in expert-rated reasoning quality on held-out cases. Secondary measures include successful knowledge transfer, harmful overgeneralization, feedback richness, task time, workload, and perceived control. Mixed-effects analysis compares learning gains across feedback conditions while accounting for participant and case differences. Qualitative coding examines the tacit knowledge captured through each approach, including account-specific exceptions, stakeholder relationships, temporal patterns, and general heuristics. By connecting knowledge types and memory characteristics to successful and failed transfers, the analysis explains when conversational elicitation offers advantages over simpler feedback. Success requires not merely richer feedback, but demonstrably better reasoning on new cases, appropriate application boundaries, and transparent, human-governed memory.

Expected contribution

Study 2 develops a human-authorized conversational expertise-learning agent that transforms professional feedback into structured, reusable expert memory. It provides empirical evidence on when conversational elicitation offers advantages over simpler feedback mechanisms, and how scoped memory representation and selective retrieval support knowledge transfer across cases without harmful overgeneralization. The findings establish design principles for building adaptive agents that progressively learn from professional expertise while preserving transparency, contextual boundaries, and human control. The resulting expert memory can also inform the reasoning audits in Study 3 and evidence-grounded workflows in Study 4.

Study 3 — Auditing Human–AI Reasoning in Evidence-Based Professional Judgment

RQ3 — Where, why, and with what consequences do AI systems and experienced professionals diverge in evidence selection, interpretation, counterevidence, and action reasoning at the same pivotal moment?

Research purpose

AI systems are increasingly capable of producing plausible professional judgments, yet agreement on final predictions does not necessarily imply agreement on how evidence is selected, interpreted, or translated into action. Study 3 investigates the underlying reasoning differences between AI and experienced professionals, treating disagreement as a subject of inquiry rather than simply an error to eliminate. The study aims to distinguish acceptable differences in professional judgment from systematic AI blind spots and potentially complementary perspectives. Unlike Study 2, which investigates how agents learn from expert feedback, Study 3 develops an independent framework for auditing the quality, structure, and consequences of human–AI reasoning differences.

Structured reasoning protocol

Using pivotal moments identified in Study 1, experienced professionals and AI independently assess identical, de-identified evidence under a shared protocol. Each produces a structured reasoning record covering: (1) five prioritized evidence items; (2) contextual interpretation of each item; (3) expected near-term trajectory and confidence; (4) strongest counterevidence or alternative explanation; (5) recommended next action; and (6) what additional evidence would change the assessment. Professionals complete their assessments before viewing AI outputs, while the AI operates with the same evidence and task instructions. This enables direct comparison of reasoning processes rather than relying solely on prediction accuracy or final recommendation agreement.

Reasoning dimensions and mismatch taxonomy

The audit examines four interconnected dimensions: evidence selection, identifying which information receives attention; contextual interpretation, including temporal changes and stakeholder relationships; counterevidence and uncertainty, assessing alternative explanations and confidence; and action reasoning, examining whether recommendations logically follow from the available evidence. Potential mismatches include overlooking changes in stakeholder authority, confusing current states with temporal trends, omitting contradictory evidence, expressing unjustified certainty, or recommending actions unsupported by the case context. Importantly, the study also examines cases where AI identifies relevant evidence or alternative interpretations overlooked by individual professionals, recognizing the possibility of complementary reasoning rather than assuming human infallibility.

Evaluation

Evaluation proceeds in two complementary stages.

First, reasoning divergence and quality are assessed using approximately 10–15 experienced SEs and sales managers, who independently evaluate overlapping pivotal-moment cases alongside a fixed AI system. All participants receive identical evidence and complete the structured reasoning protocol without seeing AI outputs. Human–human comparisons establish a reference for normal professional disagreement, while human–AI comparisons identify distinctive reasoning differences. Evidence selection is evaluated through overlap and ranking metrics, and independent expert ratings assess contextual interpretation, counterevidence, uncertainty, and action feasibility using a common rubric.

Second, reasoning consequentiality is evaluated by a separate group of approximately 10–15 experienced sales professionals who did not participate in the initial assessments. Reviewers receive identical deal contexts but are randomly assigned one of two anonymized evidence interpretations drawn from AI–human disagreements; matched human–human alternatives serve as comparison cases. Each reviewer independently records their deal assessment, evidence-seeking priorities, and recommended next actions without seeing the alternative interpretation. Comparing resulting decisions across assignments tests whether differences in reasoning meaningfully influence professional judgment rather than merely appearing different in wording. Case assignment is counterbalanced across reviewers; final sample sizes will be informed by pilot-based power analysis.

Analysis

Quantitative analysis first compares human–AI reasoning divergence against human–human variability across evidence selection, contextual interpretation, counterevidence, and action reasoning, accounting for participant and case differences. The second analysis compares deal assessments, evidence-seeking priorities, and recommended actions across randomly assigned interpretations, including matched human–human comparisons. It estimates which types of divergence lead to consequential shifts in downstream judgments rather than relying on direct similarity ratings.

Qualitative analysis investigates representative cases of high and low reasoning divergence and decision consequentiality. Thematic coding identifies recurring patterns, including overlooked stakeholder relationships, misinterpreted temporal changes, omitted counterevidence, unjustified certainty, and potentially complementary interpretations. Integrating both analyses distinguishes superficial disagreement from consequential reasoning differences, explains why some small interpretive differences substantially affect professional decisions, and identifies where AI reasoning presents limitations or complementary value. These findings support an empirically grounded taxonomy of human–AI reasoning divergence and inform subsequent agent design.

Expected contribution

Study 3 establishes a human-centered reasoning audit framework that examines how AI and professionals interpret the same evidence and whether their reasoning differences meaningfully affect professional decisions. By comparing human–AI divergence against human–human variability and evaluating its impact on deal assessments, evidence-seeking priorities, and recommended actions, the study distinguishes superficial disagreements from consequential reasoning differences. The findings provide an empirically grounded taxonomy of AI reasoning limitations and complementary strengths, along with design principles for supporting evidence-based human–AI judgment. These insights can inform expertise learning in Study 2 and the design of reliable, human-controlled agents in Study 4.

Study 4 — Building an Evidence-to-Action Agent for Professional Work

RQ4 — How can an evidence-grounded agent transform validated professional judgments into context-specific strategies and actionable work artifacts while preserving provenance, organizational constraints, and human control?

Research purpose

Studies 1–3 establish how AI can identify consequential evidence shifts, learn from professional expertise, and audit reasoning differences between humans and AI. Study 4 investigates how these capabilities can support forward-looking professional action. Rather than merely explaining historical deal risks, the agent helps SEs determine what to investigate, which risks to address, and how to strengthen future engagement. The central question is whether integrating evidence-grounded reasoning with agentic planning and artifact generation can improve the quality, actionability, and contextual appropriateness of professional strategies. The goal is to support decisions that may improve deal outcomes while keeping professionals responsible for final judgments and actions.

Evidence-to-action agent design

The agent integrates three capabilities: contextual reasoning, strategic planning, and verified artifact generation. It retrieves relevant deal evidence, expert-approved memories, and organizational knowledge to assess the situation and formulate next steps. When information is missing or conflicting, it asks targeted clarification questions before proposing strategies. It then generates context-specific artifacts while verifying factual claims, evidence provenance, and organizational constraints. Professionals can inspect supporting evidence, revise recommendations, and approve final outputs, ensuring that forward-looking actions remain grounded in validated judgments.

Artifact generation and workflows

The prototype supports two professional tasks: (1) a next-meeting strategy brief that synthesizes deal risks, stakeholder concerns, competing interpretations, and opportunities into prioritized questions, mitigation strategies, and actionable next steps; and (2) an executive account-review slide that communicates deal trajectory, critical evidence, strategic recommendations, and decisions requiring management attention. Both artifacts are designed to translate retrospective evidence into forward-looking professional action, tailored to different audiences and decision-making needs.

The system combines LLM APIs with structured prompting, tool-based retrieval, and expert memory from Study 2. An agentic planning component selects relevant evidence, identifies information gaps, and organizes recommendations according to the artifact's purpose and audience. Structured output schemas and approved templates guide content organization and visual hierarchy, while configurable brand rules and guardrails enforce terminology, tone, confidentiality, and organizational policies. A claim-level verification mechanism distinguishes source-supported facts from AI interpretations and proposed actions, links key claims to their provenance, and flags unsupported or conflicting

information for clarification. Professionals can inspect the evidence-to-recommendation mapping, revise content and presentation, and approve final artifacts before use.

Evaluation

A controlled within-subject study with approximately 15–25 experienced SEs and sales managers, subject to pilot-based power analysis, compares three conditions using matched deal cases: (A) one-shot LLM generation, (B) template-grounded generation with organizational rules, and (C) the full evidence-to-action agent with retrieval, expert memory, clarification, planning, and verification. Participants create both artifacts under counterbalanced conditions. Blinded expert reviewers assess evidence fidelity, risk coverage, actionability, template compliance, and brand consistency. Additional measures include unsupported claims, revision effort, task time, workload, and perceived control.

A complementary audience evaluation examines how effectively the artifacts communicate professional judgments. Independent SEs and managers evaluate meeting briefs, while sales directors and senior managers assess executive slides. Blinded reviewers rate information clarity, narrative flow, visual hierarchy, audience appropriateness, and decision usefulness. Short comprehension tasks measure whether readers can correctly identify key risks, supporting evidence, and recommended actions.

Analysis

Quantitative analysis uses mixed-effects regression to compare conditions while accounting for participant, case, artifact type, and reviewer differences. A vs. B comparisons test improvements in layout clarity, information hierarchy, and organizational compliance; B vs. C comparisons examine additional gains in evidence fidelity, risk coverage, actionability, and revision efficiency. Audience ratings, comprehension accuracy, and reading time assess whether these improvements produce clearer and more usable artifacts.

Qualitative analysis examines final artifacts, revision histories, and participant interviews to explain observed differences. For meeting briefs, thematic coding focuses on information organization, clarity of risk explanations, and logical progression from evidence to recommended actions. For executive slides, it examines visual hierarchy, layout consistency, narrative coherence, information density, and clarity of decision requests. Integrating both analyses identifies where agentic generation adds value beyond templates, where it introduces unnecessary complexity, and how professionals retain control over content and presentation.

Expected contribution

Study 4 contributes an evidence-grounded agent architecture that connects retrospective conversation intelligence with prospective professional action. It provides empirical evidence on when agentic planning, retrieval, clarification, and verification improve the accuracy, comprehensibility, and practical usefulness of professional artifacts beyond conventional LLM generation. The findings establish design principles for producing traceable, audience-appropriate, and actionable work artifacts under human control, while providing a foundation for future longitudinal research on professional decisions and downstream business outcomes.

Research roadmap

The four studies form an evidence-to-action research program with distinct questions and evaluation outcomes. Study 1 identifies turn-level evidence shifts and tests whether professionals regard them as consequential. Study 2 evaluates whether conversational feedback produces scoped expert memory that improves reasoning on new cases without harmful transfer. Study 3 independently audits how humans and AI reason from the same evidence, and tests whether disagreements actually change assessments, information-seeking, or next actions. Study 4 translates validated evidence and expert knowledge into meeting briefs and executive slides, evaluating both creator experience and audience comprehension. The studies inform one another but are not dependent on a single shared success criterion: an unsuccessful memory transfer or an unresolved reasoning disagreement is itself a finding that can guide safer artifact generation.

Study	Primary analytic object	Concrete output	Feeds into
1	Turn-level evidence changes	Validated detection + evidence-shift taxonomy	Cases for expertise elicitation and independent audit
2	Expert teaching and memory transfer	Human-approved, scoped memory + transfer limits	Reusable context for future agent workflows
3	Human–AI reasoning consequences	Human–human benchmark + consequential divergence taxonomy	Evidence and uncertainty controls for agent design
4	Evidence-to-action artifact use	Verified briefs/slides + creator and audience findings	Design principles and future field evaluation

Research Vision and Broader Impact

This research agenda begins not with a hypothetical AI application, but with DealLens—an implemented, empirically evaluated foundation that provides real-world evidence, an established analytical workflow, and a concrete setting for investigating human–AI collaboration. Building on this foundation, the proposed studies offer distinct yet interconnected

contributions, sharing reusable evidence pipelines, agent infrastructure, evaluation protocols, and professional tasks. This structure supports focused empirical investigations while enabling a coherent, cumulative PhD research program.

Importantly, DealLens is a research testbed, not the boundary of the research. The underlying challenges—interpreting evolving evidence, learning situated human expertise, understanding human–AI reasoning differences, and translating judgments into meaningful support—extend well beyond sales. These questions are particularly relevant to mental health and well-being support, healthcare coordination, education, and collaborative knowledge work, where effective AI assistance requires contextual sensitivity, longitudinal understanding, calibrated trust, and respect for human agency. While each domain introduces distinct ethical and practical constraints, the proposed methods offer a foundation for investigating how memory-augmented, evidence-grounded agents can adapt to human needs without overstepping their roles.

Ultimately, the vision is to advance AI that does not replace human judgment, but becomes more capable through interaction with it, helping people recognize consequential evidence, reason through uncertainty, learn from experience, and turn informed judgments into meaningful action.

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